Predicting the Unpredictable: SIR Models and Uncertainty Quantification in COVID-19

Models of pandemics: What should we do differently next time?

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COVID-19

- In the **January 2020** a **new virus** emerges in **Wuhan**
- **Scientists** seek a way to **simulate** the **evolution** of the **epidemic**
- **How fast** will it **spread**? **How many** people will get **infected**? **How can** the **healthcare system** handle it? **How to** evaluate possible **interventions**?

Compartmental models

- **Different methods** exist to **describe** the **epidemic dynamics**
- **Compartmental models** are mathematical models used to **simulate** the **spread** of **infectious diseases**
- The **most popular** model is the **Susceptible-Infectious-Resolved** (SIR) model
- **Compartmental models** assume that at **each time point population** is divided in **groups** which are **homogeneous** in respect to the **health status** of the **individuals**, and that **transitions** between **compartments** follow **simple probabilistic rules**

A bit of history

- The **SIR model** was introduced by Kermack and McKendrick (1927)
- They **introduced** an **innovative approach** based on
  - **compartmentalization** of the population

$$S \xrightarrow{\beta} I \xrightarrow{\gamma} R$$

  - **difference equations** to model infection spread

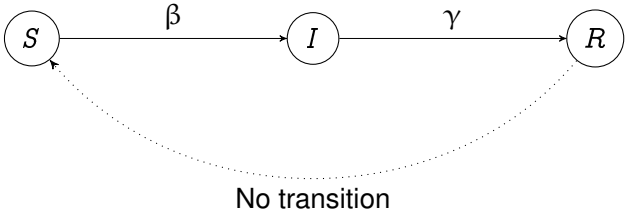
$$\begin{cases} S(t) = & S(t-1) - \beta \frac{I(t-1)}{S(0)} S(t-1) \\ I(t) = & I(t-1) + \beta \frac{I(t-1)}{S(0)} S(t-1) - \gamma I(t-1) \\ R(t) = & R(t-1) + \gamma I(t-1) \end{cases}$$

SIR model parameters

- $\gamma = \frac{1}{T}$ is the **resolution probability**, where T is the **average time** spent by an **infected individual** in I
- β is the **probability of transmission** of the **infection** and **it regulates** the **transitions** from S to I
  - The **ability** of a **pathogen** to **spread**
- $\beta \frac{I(t)}{S(0)}$ is the **infection probability** for **susceptible** at time t and **it depends** on the **number of circulating infections** at the same time
  - **Probability** of a **person**, exposed to a **pathogen**, developing the **disease**
- We use **optimization algorithm** to **find** the **parameters** that **minimize** the difference between **generated** and **observed** data

SIR model assumptions

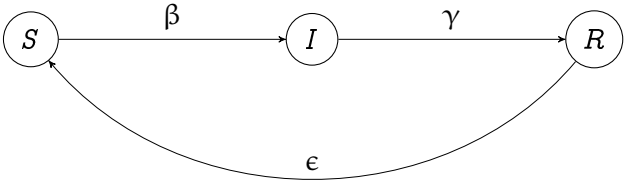
- **Starting** with a **population** of **dimension** N with a fixed number of **infected** and **recovered**
- **Closed population** without demographic turnover
- **Homogeneously** mixed population
- The **re-infection probability** is equal to **0**



- We can make **changes** to **extend** the **model** but we obtain **more complex models**

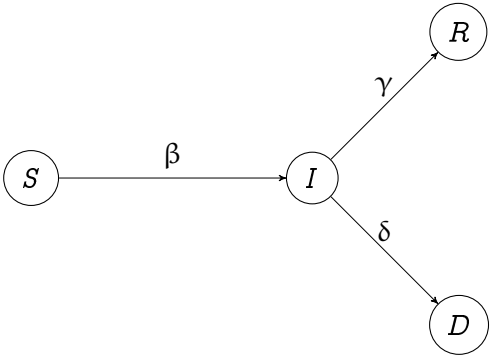
SIR model extensions

- **Susceptible-Infectious-Resolved-Susceptible** (SIRS) model



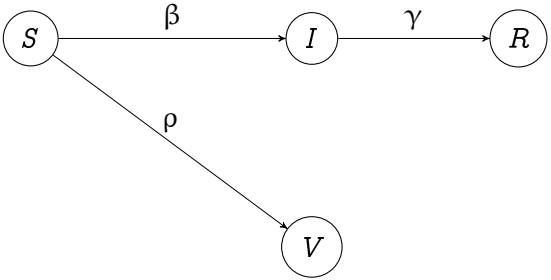
SIR model extensions

- **Susceptible-Infectious-Resolved-Deceased** (SIRD) model



SIR model extensions

- **Susceptible-Infectious-Resolved-Vaccinated** (SIRV) model



SIR model extensions

- **Susceptible-Exposed-Infectious-Resolved** (SEIR) model

$$S \xrightarrow{\mu} E \xrightarrow{\beta} I \xrightarrow{\gamma} R$$

Key metrics

- $R_0 = \frac{\beta}{\gamma} = \beta T$ is the **basic reproduction number** i.e. the **average number** of people infected by the **first infected person** at time **0**
- $R_t = R_0 \frac{S(t)}{S(0)}$ is the **effective reproduction number** i.e. the **average number** of people infected by one **infected person** at time t

Example

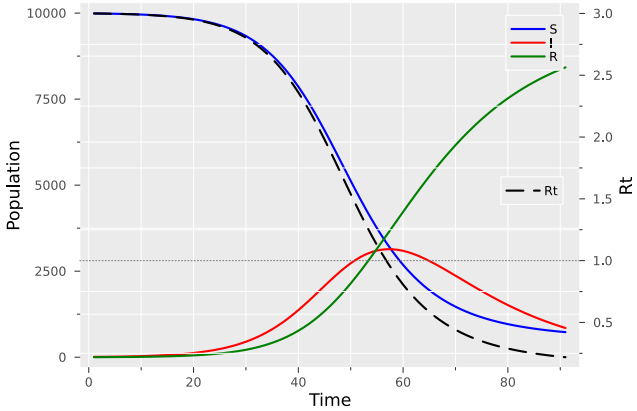


Figure: SIR parameters $T = 14$ and $R_0 = 3$ under no interventions

Modeling the COVID-19 in Italy

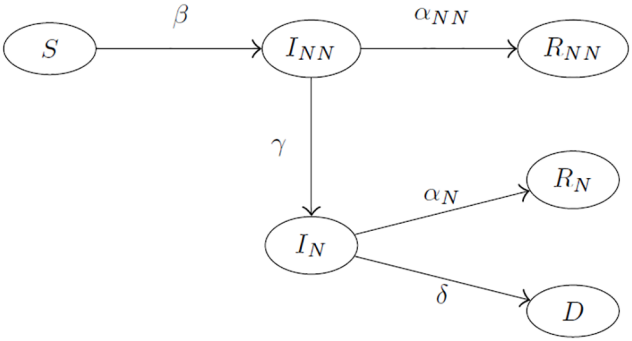


Figure: Extension of SIR model developed by Baccini et al. (2021)

Modeling the COVID-19 in Italy

$$
\begin{cases}
S(t) = S(t-1) - \beta \frac{S(t-1)}{S(0)} I_{NN}(t-1) \\
I_{NN}(t) = I_{NN}(t-1) + \beta \frac{S(t-1)}{S(0)} I_{NN}(t-1) - \gamma I_{NN}(t-1) - \alpha_{NN} I_{NN}(t-1) \\
I_N(t) = I_N(t-1) + \gamma I_{NN}(t-1) - \alpha_N I_N(t-1) - \delta I_N(t-1) \\
R_{NN}(t) = R_{NN}(t-1) + \alpha_{NN} I_{NN}(t-1) \\
R_N(t) = R_N(t-1) + \alpha_N I_N(t-1) \\
D(t) = D(t-1) + \delta I_N(t-1)
\end{cases}
$$

Global Sensitivity Analysis

- The study of **how the uncertainty** in the **output of a model** can be divided and allocated to **different sources of uncertainty** in its **input** (Saltelli et al., 2008)
- **Sobol' sensitivity indices** decompose the **variance** of the **model output** into contributions due to **input factors** using **index defined in** $[0, 1]$
- **First-order index:** Measures the **importance of a single factor** as the expected reduction in variance if it **were fixed**
- **Total-order index:** Measures the **total importance of a factor**, considering both its direct effect and all interactions with **other factors**

GSA implications

- **GSA** is **useful to identify** the **most influential parameters**
  - **Additional evidence** on **transition times** is required
  - In particular for the **time** from **infection** to **recovery** for **not-notified individuals**
  - **Not-notified individuals** are the **only ones able** to **spread the contagion**
- **What we need?** A **better characterization** of the **infection** in **asymptomatic**

Policy implications

- **Background**
  - **First wave** of COVID-19 (March - April 2020)
  - Model **predicts epidemic** under varying **interventions**
- **Findings**
  - Limited testing + Light quarantine ~**70,000 deaths**
  - Mass testing + Strict quarantine ~**30,000 deaths**
- **Implications**
  - **Early interventions** are crucial
  - Combo of **quarantine** and **testing** is the **key** to **control**

Limitations

- **Homogeneously** mixed population
- No **demographic** changes
- **Constant** transition parameters
- **Possible** alternative in literature?

Conclusions

- **SIR models** offer a **conceptual framework** to describe **disease dynamics**
- **Simple**, **fast**, and **interpretable**
- Can be used even **with limited data**
- They **help estimate key metrics** supporting **public health decisions**

References

Baccini, M., Cereda, G., and Viscardi, C. (2021). The first wave of the sars-cov-2 epidemic in tuscany (italy): A si2r2d compartmental model with uncertainty evaluation. *PLoS One*, 16(4):e0250029.

Kermack, W. O. and McKendrick, A. G. (1927). A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, 115(772):700–721.

Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., and Tarantola, S. (2008). *Global sensitivity analysis: the primer*. John Wiley & Sons.

Thank you for your attention